

**GREAT LAKES UNIVERSITY
UNIVERSITY OF KISUMU
COURSE OUTLINE**

SCHOOL: Community Health & Development
DEPARTMENT:
CAMPUS/SEMESTER: Kisumu Campus/January-April 2026

COURSE CODE/TITLE: : BIostatistics

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COURSE DESCRIPTION: Basic statistical measurements of central tendency and dispersion, means, median, hypothesis testing, estimation, confidence intervals, correlation and regression analysis; Probability and Inference. Theoretical Distributions: The principles, assumptions, on Normal, Binomial and Poisson distributions. Sampling techniques: Determination of sample size: Sample size estimation for various types of studies; case-control, evaluative, cross-sectional, prospective, clinical trials studies. Hypothesis Testing: Null Hypothesis, Alternative Hypothesis: Level of confidence; Type I and Type II errors; Power of the Test; Confidence intervals; Testing the difference between two sample means and proportions; Chi-Square and Student t-test, McNamar's test. Regression: Multiple regressions, logistic regression, fitting of model to data, confidence intervals for regression line, interpretation of regression outputs, meaning of R^2 Pearson coefficient. Analysis of variance: (ANOVA) Two-by-two tables, multi tables, assumptions, analysis, and interpretation of ANOVA tests.

ELOs: By the end of the course the learner will:

- i) Enhance understanding of statistical measures and inference through practical application and analysis.
- ii) Acquire a comprehensive knowledge of theoretical distributions through the study and application of their properties, characteristics, and statistical inference techniques.
- iii) Develop proficiency in sampling techniques and the determination of sample size, demonstrated through the

application of appropriate sampling methods and accurate sample size calculations in practical research scenarios.

- iv) Attain competence in hypothesis testing, regression analysis, and analysis of variance (ANOVA) through the successful application of these statistical methods in data analysis, interpretation of results, and drawing meaningful conclusions based on statistical inference.

Week	Topic	ELOs	Sub-Topic
1	Introduction to Biostatistics and Data Description	- Describe the basic principles of biostatistics and its importance in healthcare research. - Apply appropriate measures of central tendency and dispersion to summarize and interpret biomedical data. - Describe different types of data and variables encountered in biostatistical analysis.	- Types of data and variables - Descriptive statistics: measures of central tendency and dispersion - Calculation and interpretation of means, medians, and measures of variability
2	Probability and Theoretical Distributions	- Apply probability concepts and rules to calculate probabilities in the context of healthcare data. - Understand the principles and assumptions underlying the Normal, Binomial, and Poisson distributions in biostatistics. - Apply the properties of these distributions to model and analyze healthcare data.	- Basic concepts of probability theory - Probability rules and calculations - Probability distributions: Binomial, Poisson, Normal, t-, F- and Chi-square
3	Sampling and Sampling Techniques	- Differentiate between Probability and Non-probability sampling approaches, understand their advantages and disadvantages, and apply appropriate sampling techniques based on the research context. - Explain the concept of sampling distributions and their role in statistical inference. - Apply the central limit theorem to estimate population parameters and make inferences about healthcare datasets.	- Probability and Non-Probability Sampling Techniques - Sampling distributions and the central limit theorem
4	Confidence Intervals	- Construct and interpret confidence intervals for means, proportions, and other parameters in biostatistics. - Determine appropriate sample sizes for different types of studies in healthcare research. - Evaluate the precision and reliability of estimates using confidence intervals in the context of clinical trials and epidemiological studies.	- Construction and interpretation of confidence intervals - Confidence intervals for means and proportions - Sample size determination for different types of studies - Case-control, evaluative, cross-sectional, prospective, and clinical trials studies
5	Hypothesis Testing	- Formulate null and alternative hypotheses for research questions in biostatistics. - Calculate and interpret p-values and make decisions based on hypothesis tests in the context of healthcare studies. - Understand the concepts of Type I and Type II errors and their implications in hypothesis testing in biostatistics.	- Null and Alternative hypotheses - Type I and Type II errors - Significance level, p-value, and critical regions - One-sample and two-sample hypothesis tests - Student's t-test and its applications - Chi-Square test for independence
6	ANOVA	Understand the concept and assumptions of ANOVA and its applications in healthcare research.	- Introduction to ANOVA - One-way ANOVA: assumptions, calculations, and interpretation.

		2. Perform one-way ANOVA to compare means of multiple groups in biomedical studies. 3. Interpret ANOVA results, including F-statistics and p-values, and conduct post-hoc tests for multiple comparisons.	• Two-way ANOVA and multi-way ANOVA. • Post-hoc tests and multiple comparisons
7	Non-Parametric Tests	1. Identify situations where non-parametric tests are applicable in biostatistical analysis. 2. Apply non-parametric tests such as the Wilcoxon signed-rank test, Mann-Whitney U test, and Kruskal-Wallis test to healthcare datasets. 3. Interpret non-parametric test results and understand their advantages and limitations compared to parametric tests.	• Introduction to non-parametric tests ◦ Wilcoxon signed-rank test ◦ Mann-Whitney U test ◦ Kruskal-Wallis test ◦ McNemar test
8	Correlation and Regression Analysis	1. Analyze and interpret the strength and direction of relationships between variables in healthcare datasets using correlation coefficients. 2. Apply simple linear regression analysis to model and predict health outcomes based on explanatory variables. 3. Interpret regression outputs, including coefficients, p-values, and R-squared, in the context of biomedical research.	• Understanding the relationship between variables • Calculation and interpretation of correlation coefficients • Simple linear regression analysis • Fitting a regression model to data • Confidence intervals and hypothesis testing for regression coefficients • Introduction to logistic regression
9	Advanced Topics in Statistical Inference	1. Understand the concept of statistical power and its importance in designing and interpreting healthcare studies. 2. Determine sample sizes for hypothesis testing and power calculations in biostatistics. 3. Apply meta-analysis techniques to combine and analyze results from multiple studies in healthcare research.	• Power of the test • Sample size determination for hypothesis testing • Non-inferiority and equivalence testing • Meta-analysis: combining results from multiple studies
10	Review and Applications	1. Review and consolidate key concepts and techniques covered throughout the course. 2. Apply statistical methods and reasoning to real-world scenarios in biostatistics. 3. Demonstrate critical thinking and ethical considerations in statistical analysis and data interpretation in the healthcare field.	• Recap of key concepts and techniques covered in the course • Application of statistical methods to real-world scenarios • Ethical considerations in statistical analysis

ASSESSMENT STRATEGY: Each week will consist of lectures, practical examples, and problem-solving sessions.
Assignments and quizzes will be given to reinforce understanding and assess learning progress.
End of Semester Examination

REFERENCES:

1. Bernard Rosner (2016) Fundamentals of Biostatistics 8th Edition. CENGAGE Learning. ISBN: 978-1-305-26892-0
2. Chang, Mark. (2013). Modern Issues and Methods in Biostatistics (Statistics for Biology and Health).

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How to determine the correct sample size of a research

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Sample size calculation

- The sample size calculation or estimation has no one single formula that can apply universally to all situations and circumstances.
- The sample size estimation can be done either by using;
 - Manual calculation,
 - Sample size software,
 - Sample size tables from scientific published articles,
 - Adopting various acceptable rule-of-thumbs.

Factors that must be estimated to calculate sample size

- There are four factors that must be known or estimated to calculate sample size:
 1. The effect size;
 2. The population standard deviation;
 3. The power of the experiment; and
 4. The significance level.

1. The effect size

- The effect size indicates how meaningful the relationship between variables or the difference between groups is. It also indicates the practical significance of a research outcome.
- The effect sizes are independent of the sample size, only the data is used to calculate effect sizes.
- A large effect size means that a research finding has practical significance, while a small effect size indicates limited practical applications.
- The practical significance shows that the effect is large enough to be meaningful in the real world.

1. The effect size cont'd

- The most common effect sizes are Cohen's d and Pearson's (r).
- **Cohen's (d)** measures the size of the difference between two groups.
- **Pearson's (r)** measures the strength of the relationship between two variables. It is better to use a statistical software to calculate Pearson's r accurately from the raw data.

Cohen's d formula

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s}$$

where,

- $\bar{x}_1$ = mean of Group 1
- $\bar{x}_2$ = mean of Group 2
- s = standard deviation

Pearson's r formula

$$r = \frac{n \sum xy - (\sum x)(\sum y)}{\sqrt{[n \sum x^2 - (\sum x)^2][n \sum y^2 - (\sum y)^2]}}$$

where,

- r_{xy} = strength of the correlation between variables x and y
- n = sample size
- $\sum$ = sum of what follows
- X = every x-variable value
- Y = every y-variable value
- XY = the product of each x-variable score times the corresponding y-variable score

Estimation of the effect size

- It is usually quite challenging to estimate an accurate effect size since the exact value of the effect size is not known until the study is completed. However, the researcher will still have to set the value of effect size for the purpose of sample size calculation or estimation.

Estimation of the effect size cont'd

The effect size can be estimated from:

- **Literature review**

- the acceptable and desirable way.
- it is better to obtain the needed information from recent articles (within 5 years) that used almost similar design, same treatment and similar patient characteristics.

- **From historical data or secondary data**

- provided researcher has access to all data of different group
- not always feasible since a new intervention may not have been assessed yet

- **Educated guess or expert opinion**

- researches/experts can use his experience and knowledge to set up an effect size that is scientifically or clinically meaningful

2. The population standard deviation

- **The standard deviation is used with continuous data, but not categorical data.**
- The population standard deviation is a parameter.
- It is a fixed value calculated from every individual in the population.
- The standard deviation, is appropriate when the continuous data is not significantly skewed or has outliers.
- **It can be derived from literature review, pilot study or from any reliable source.**

The **sample standard deviation formula** is:

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

where,

s = sample standard deviation

$\sum$ = sum of...

$\bar{X}$ = sample mean

n = number of scores in sample.

3. The power

- The power of study is the pre-study probability that the effect will be detected and the test will reject the null hypothesis.
- In other words, If the power is too low, there is little chance of detecting a significant difference, and non-significant results are likely even if real differences truly exist.
- **The power is arbitrarily set to 80 or 90% which means that the study has 80% or 90% chance of having a statistically significance results.**
- **$\beta = 1 - \text{power}$** , is the chance of obtaining a false-negative result (the experiment will fail to reject an untrue null hypothesis or to detect the specified treatment effect).
- Statistics provides the following Power Analysis procedures:
 - One Sample T-Test
 - Paired Sample T-Test
 - Independent Sample T-Test
 - One-way ANOVA

3. The power cont'd

Statistical Power and Beta

- **Type I error Alpha (α):** rejecting the null hypothesis of no effect when it is actually true. **False positive**
- **Type II error Beta (β):** not rejecting the null hypothesis of no effect when it is actually false. **False negative**

	Do not reject H_0	Reject H_0
H_0 is true	Correct Decision	Incorrect Decision: Type I error α
H_0 is false	Incorrect Decision: Type II error β	Correct Decision

4. The significance level

- The significance level, also known as alpha (α) is the probability that a positive finding is due to chance alone. It is a measure of the strength of the evidence that must be present in the sample before rejecting the null hypothesis and concluding that the effect is statistically significant.
- **The researcher determines the significance level before conducting the experiment and is usually chosen to be 0.05 or 0.01.**
- That is, the investigator wishes the chance of mistakenly designating a difference “significant” (when in fact there is no difference) to be no more than 5 or 1%.
- Significance level is correlated with power: increasing the significance level (from 5% to 10%) increases power.

Steps to follow in sample size determination

1. Understand the Objective of the Study

The objective of a study has to be measurable or in other words, can be determined by using statistical analysis.



2. Select the Appropriate Statistical Analysis

The Researchers have to decide the appropriate analysis or statistical test to be used to answer the study objective. The formula that will be used to estimate or calculate the sample size will be the same formula for performing the statistical test that will be used to answer the objective of study.



3. Calculate or Estimate the Sample Size

Estimating or calculating the sample size can be done either by using manual calculation, sample size software, sample size tables from scientific published articles, or by adopting various acceptable rule-of-thumbs.



4. Provide an additional allowance during Subject Recruitment to Cater for a Certain Proportion of Non-Response

A minimum required sample size means the minimum number of subjects a study must have after recruitment is completed. Therefore, the researchers must ideally be able to recruit subjects at least beyond the minimum required sample size. It is advisable to add 20-30% more. If the chance of non response is high then it can be increased up to 40-50%.



5. Write a Sample Size Statement by outlining steps from 1 to 4

The sample size statement is usually included in the protocol or manuscript. There are various styles to write the statement.

Examples of free sample size calculator tools

- **ClinCalc LLC. Sample Size Calculator:** <https://clincalc.com/stats/SampleSize.aspx>
 - A free online sample size calculator
- **Epi Info™** is a free software that can be downloaded from the Centers for Disease Control and Prevention (CDC) website at: <https://www.cdc.gov/epiinfo>.
 - Watch the [Epi Info™ 7 Tutorial Videos](#).
- **OpenEpi.com:** <https://www.openepi.com>
 - An open-source web tool that provides epidemiologic statistics.
- **PS: Power and Sample Size Calculation:** <https://biostat.app.vumc.org/wiki/Main/PowerSampleSize>
 - A free interactive program for performing power and sample size calculations
- **StatCalc:** <https://www.cdc.gov/epiinfo/user-guide/statcalc/statcalcintro.html>
 - A utility tool in [Epi Info™](#) and statistical calculator that produces summary epidemiologic information.
 - Six types of calculations are available including Sample Size and Power calculations for Population Survey, Cohort or Cross-Sectional, and Unmatched Case-Control.
- **STEPS [Sample Size Calculator](#) and [Sampling Spreadsheet](#)**

Sample size for dichotomous variables

- **Dichotomous variable** is often expressed as a rate or proportion of a yes/no outcome, like the occurrence of disease or survival at a given time.
- Usually the aim of the experiment is to compare the proportions in two groups.
- In such a case, the sample size, n , can be given by the following formula:

$$n = C \frac{p_c q_c + p_e q_e}{d^2} + \frac{2}{d} + 2$$

$$q_c = 1 - p_c$$

$$q_e = 1 - p_e$$

$$d = |P_c - P_e|$$

d is the difference between p_c and p_e .

C is a constant that depends on the values chosen for α and β

P_c = proportion in control group

P_e = proportion in experimental group

r_c = the number of events in control group or group c

N_c = the total number of animals in control group or group c ,

r_e = the number of events in the experimental group or group e

N_e = total number of animals in the experimental group or group e

$$p_c = \frac{r_c}{N_c}; p_e = \frac{r_e}{N_e},$$

Example 1 sample size for dichotomous variables

Suppose previous data suggest that the spontaneous incidence of uterine tumor in cows is 50% and, in an experiment, to determine whether chemical substance in the fodder increases the incidence of this tumor, studying the same type of cows, the scientist specifies that if the incidence is 60%, he/she would like to have a 90% chance of detecting this increase. Testing at $p = 0.05$, the sample size is estimated as follows:

$$n = C \frac{p_c q_c + p_e q_e}{d^2} + \frac{2}{d} + 2$$

$p_c=0.5$, $p_e=0.6$, $d=0.1$, For $\alpha = 0.05$ and $1-\beta = 0.9$ so $C = 10.51$.

$$N = 10.51 * (0.5 * 0.5 + 0.6 * 0.4 / 0.01) + 2 / 0.1 + 2 = 537$$

So, for the study the researcher will need around 537 cow in each group.

Sample size for dichotomous variables cont'd.

- For dichotomous variables in a single population, the formula for determining sample size is:

$$n = p \left(1 - p \right) \left(\frac{Z}{E} \right)^2$$

- Z = the standard normal distribution reflecting the confidence level that will be used and it is usually set to Z = 1.96 for 95%.
- E = the desired margin of error.
- p = approximate anticipated proportion of successes in the population and it usually ranges between 0-1. If unknown, the value 0.5 is used to estimate the sample size.

Example 2 sample size for dichotomous variables

- A researcher wants to estimate the prevalence of ovarian cancer among women who are between 50 and 55 years of age living in his region. Suppose that the National data suggest that 1 in 400 women are diagnosed with ovarian cancer by age 55. This gives a proportion of 0.0025 (0.25%) or a prevalence of 25 per 10,000 women. If the investigator wants the estimate to be within 10 per 10,000 women with 95% confidence. The sample size is computed as follows:

Using the following formula:

$$n = p(1 - p) \left(\frac{Z}{E} \right)^2$$

$$N = 0.0025(1 - 0.0025)(1.96/0.0010)^2 = 9579.99 \quad * Z \text{ for } 95\% = 1.96$$

So roughly the sample size will be around 9580 woman between 50-55 years.

Sample size for continuous variables

- Continuous variables like measuring the concentration of a substance in a body fluid or a physiological function such as blood flow rate or urine output.
- It is often critical to detect the difference in the mean of a variable between two groups, therefore the statistical analytical models may be complex.
- In case of Continuous data, the sample size can be given by the following formula

$$n = 1 + 2C \left(\frac{s}{d} \right)^2$$

(s) = population standard deviation of the variable.

(d) = the magnitude of the difference the investigator wishes to detect, often called the effect.

(C) = a constant dependent on the selected value of α and β

Example sample size for continuous variables

- Lets suppose that the concentration of toxic metal in body fluid in a group of individuals is 9mmol/l, with a standard deviation of 5 mmol/l, and that a new substance discovered reduces the concentration of this metal in the body fluid is to be tested to know whether it alters the level of the metal in the body fluid. If the scientist would like to be able to detect a 3mmol reduction of the metal in body fluid between control and treated persons with a power of 90% and a significance level of 5%, using a two-tailed unpaired t-test (two-tailed because the tested substance might increase the level of the metal).

Using the following formula:

$$n = 1 + 2C\left(\frac{s}{d}\right)^2$$

C can be determined from a statistical table, which gives values for C .

For $\alpha = 0.05$ and $1-\beta = 0.9$, C is 10.51

$$N = 1 + 21(5/3)^2 = 59$$

So the sample size needed is roughly around 59 for each group and 118 for the whole study.

Sample size for time to an event

- The statistical analysis of time to an event involves complicated statistical models.
- Two simple approaches to estimating sample size for this type of variable
- **The first approach:** Is using the proportions in the two experimental groups exhibiting the event by a certain time to estimate the sample size.
- This method converts time to an event into a dichotomous variable.
- The sample size is estimated by the following formula:

$$n = C \frac{p_c q_c + p_e q_e}{d^2} + \frac{2}{d} + 2$$

$$q_c = 1 - p_c$$

$$q_e = 1 - p_e$$

$$d = | p_c - p_e |$$

$$p_c = \frac{r_c}{N_c}; p_e = \frac{r_e}{N_e}$$

d is the difference between p_c and p_e (expressed as a positive quantity)

C is a constant that depends on the values chosen for α and β

r_c = the number of events in control group or group c

N_c = the total number of animals in control group or group c ,

r_e = the number of events in the experimental group or group e

N_e = total number of animals in the experimental group or group e

Sample size for time to an event cont'd

- **The second approach:** is to treat time to occurrence as a continuous variable.
- This approach is applicable only if all tested individuals or animals are followed to event occurrence (e.g., until death or time to exhibit a disease such as cancer), but it cannot be used if some of the individuals or animals do not reach the event during the study.
- Sample size may be computed by the following:

$$n = 1 + 2C \left(\frac{s}{d} \right)^2$$

(s) = population standard deviation of the variable.

(d) = the magnitude of the difference the investigator wishes to detect, often called the effect.

(C) = a constant dependent on the selected value of α and β

References for more reading

1. Bhandari P. What is Effect Size and Why Does It Matter? (Examples) [Internet]. Scribbr. 2020 [cited 2022 Sep 9]. Available from: <https://www.scribbr.com/statistics/effect-size/>
2. Bujang MA. A Step-by-Step Process on Sample Size Determination for Medical Research. Malays J Med Sci MJMS. 2021 Apr;28(2):15–27.
3. Dell RB, Holleran S, Ramakrishnan R. Sample Size Determination. Ilar J. 2002;43(4):207–13.
4. Greenland S, Senn SJ, Rothman KJ, Carlin JB, Poole C, Goodman SN, et al. Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations. Eur J Epidemiol. 2016;31:337–50.
5. Sullivan L. Power and Sample Size Determination [Internet]. [cited 2022 Sep 9]. Available from: https://sphweb.bumc.bu.edu/otlt/mph-modules/bs/bs704_power/bs704_power_print.html
6. Standard Deviation | How and when to use the Sample and Population Standard Deviation - A measure of spread | Laerd Statistics [Internet]. [cited 2022 Sep 9]. Available from: <https://statistics.laerd.com/statistical-guides/measures-of-spread-standard-deviation.php>
7. Statistical Power: What it is, How to Calculate it [Internet]. Statistics How To. [cited 2022 Sep 5]. Available from: <https://www.statisticshowto.com/probability-and-statistics/statistics-definitions/statistical-power/>

Useful websites for sample size calculation

- ClinCalc LLC. Sample Size Calculator [website]. C2002. Available at: <https://clincalc.com/stats/SampleSize.aspx>
- Dean AG, Sullivan KM, Soe MM. OpenEpi: Open Source Epidemiologic Statistics for Public Health, Version. www.OpenEpi.com, updated 2013/04/06. Available at: <https://www.openepi.com>
- GIGAcaculator. Power & Sample Size Calculator [website]. c2017-2022. Available at: <https://www.gigacalculator.com/calculators/power-sample-size-calculator.php>
- Kohn MA, Senyak J. Sample Size Calculators [website]. UCSF CTSI. 20 December 2021. Available at <https://www.sample-size.net/>
- PS: Power and Sample Size Calculation [website]. Vanderbilt Biostatistics Wiki, c2013-2022. Available from: <https://biostat.app.vumc.org/wiki/Main/PowerSampleSize>.
- StatCalc in Epi Info™, Division of Health Informatics & Surveillance (DHIS), Center for Surveillance, Epidemiology & Laboratory Services (CSELS) [website]. CDC. Available at: <https://www.cdc.gov/epiinfo/user-guide/statcalc/samplesize.html>
- <<http://www.biomath.info>>: a simple website of the biomathematics division of the Department of Pediatrics at the College of Physicians & Surgeons at Columbia University, which implements the equations and conditions discussed in this article
- <<http://davidmlane.com/hyperstat/power.html>>: a clear and concise review of the basic principles of statistics, which includes a discussion of sample size calculations with links to sites where actual calculations can be performed
- nQuery Advisor, SPSS, MINITAB and SAS/STAT are paid statistical programs and software that can be used both for sample size calculations and statistical data analysis

HYPOTHESIS TESTING

BASIC PRINCIPLES OF HYPOTHESIS TESTING

Definition of Hypothesis Testing

- Hypothesis testing is a statistical method used to make inferences or draw conclusions about a population based on a sample of data.
 - The process involves formulating two competing hypotheses, the null hypothesis (H_0) and the alternative hypothesis (H_1 or H_A), and assessing the evidence from the sample to determine which hypothesis is more supported by the data.
- [Link to Lecture 3](#)

Null and Alternative Hypothesis

- Null Hypothesis (H_0): The null hypothesis represents the default assumption or the status quo.
 - It states that there is no significant difference or relationship between variables or that an observed effect is due to chance.
 - It is usually denoted as H_0 .
- Alternative Hypothesis (H_1 or H_A): The alternative hypothesis is the statement that contradicts the null hypothesis.
 - It represents the researcher's claim or the possibility of a significant difference or relationship between variables.
 - It is denoted as H_1 or H_A .

Examples of Null Hypotheses in Biostatistics

- These examples highlight various scenarios in biostatistics where the null hypothesis represents the absence of an effect, difference, or association between variables or populations.
 - Null Hypothesis: There is no difference in mean blood pressure between patients receiving Drug A and patients receiving a placebo.
 - Null Hypothesis: The proportion of individuals with a specific genetic mutation is the same in males and females.
 - Null Hypothesis: There is no association between smoking status and the incidence of lung cancer.

Examples of Null Hypotheses in Biostatistics

- Null Hypothesis: The mean cholesterol levels in a population before and after a dietary intervention are equal.
- Null Hypothesis: The mean body mass index (BMI) is the same across different age groups.
- Null Hypothesis: The odds of developing a certain disease are equal for individuals exposed and not exposed to a particular environmental factor.

Examples of Null Hypotheses in Biostatistics

- Null Hypothesis: There is no difference in survival rates between two different treatment groups for a specific disease.
- Null Hypothesis: The mean systolic blood pressure is the same among individuals with and without a family history of hypertension.
- Null Hypothesis: The prevalence of a certain disease is not different among individuals of different socioeconomic statuses.
- Null Hypothesis: There is no correlation between dietary intake of a specific nutrient and body weight.

Significance Level

- The significance level, often denoted as α (alpha), is the threshold set by the researcher to determine the level of evidence required to reject the null hypothesis.
 - Commonly used significance levels are 0.05 (5%) and 0.01 (1%).
 - The significance level is a measure of the strength of the evidence that must be present in your sample before rejecting the null.
 - The probability of rejecting the null when it is true.

Test Statistic

- A test statistic is a numerical value calculated from the sample data that helps in assessing the evidence against the null hypothesis.
 - The choice of test statistic depends on the nature of the hypothesis being tested and the type of data being analyzed.

Test Statistic distribution

- Under the assumption that the null hypothesis is true, the test statistic follows a known probability distribution.
- The choice of distribution depends on the type of test being conducted (e.g., t-distribution, normal distribution, chi-square distribution).
- The test statistic distribution is used to calculate the p-value.

Test Statistic distribution

- Under the assumption that the null hypothesis is true, the test statistic follows a known probability distribution.
- The choice of distribution depends on the type of test being conducted
 - **t - distribution** – for calculating CIs for and testing hypothesis about one or two means
 - **Chi-square distribution** – useful for analyzing categorical data
 - **F - distribution** – used for comparing variances and more than two means using ANOVA

p - value

- The p-value is the probability of obtaining a test statistic as extreme as or more extreme than the one observed, assuming that the null hypothesis is true.
 - It measures the strength of evidence against the null hypothesis.
 - If the p-value is smaller than the significance level (α), it is considered statistically significant, and the null hypothesis is rejected in favor of the alternative hypothesis.

Conclusion

- Based on the p-value and the significance level, a decision is made regarding the null hypothesis.
 - If the p-value is less than α , the null hypothesis is rejected, and the alternative hypothesis is supported.
 - If the p-value is greater than or equal to α , there is insufficient evidence to reject the null hypothesis.
- [Link to Lecture 4](#)

Errors in Hypothesis Testing

- There are two types of errors that can occur when making conclusions about the null hypothesis.
 - Type I Error
 - Type II Error

Type I error

- A Type I error occurs when the null hypothesis (H_0) is incorrectly rejected, even though it is true.
 - It is the error of concluding that there is a significant effect or relationship when, in reality, there is no such effect or relationship.
 - Type I error is also known as a "**false positive**."
 - The probability of committing a Type I error is equal to the chosen **significance level** (α) of the test.
 - For example, if α is set at 0.05, it means that there is a 5% chance of committing a Type I error.
- Implication:
 - Committing a Type I error can lead to the rejection of a null hypothesis that is actually true.
 - This can result in incorrect conclusions, wasted resources, and potentially adopting new practices or interventions based on faulty evidence.

Type II error

- A Type II error occurs when the null hypothesis (H_0) is incorrectly accepted, even though it is false.
 - It is the error of failing to reject the null hypothesis when there is a true effect or relationship present.
 - Type II error is also known as a "**false negative**."
 - The probability of committing a Type II error is denoted as β (beta) and is related to the power of the statistical test.
 - Power ($1 - \beta$) represents the probability of correctly rejecting a false null hypothesis.
- Implication:
 - Committing a Type II error means that a real effect or relationship goes undetected or remains unacknowledged.
 - This can lead to missed opportunities, failure to identify important findings, and a lack of action or intervention when it is necessary.

Type I and Type II error

- It's important to note that Type I and Type II errors are inversely related.
- Reducing the probability of one type of error (e.g., decreasing α to reduce Type I error) typically increases the probability of the other type of error (e.g., increasing β and thus increasing the likelihood of Type II error).

Power of a Test

- The chance of making a Type II error is denoted by β (beta);
 - its complement, $(1 - \beta)$, is the power of the test.
- The power, therefore, is the probability of rejecting the null hypothesis when it is false;
 - i.e. it is the chance (usually expressed as a percentage) of detecting, as statistically significant, a real treatment effect of a given size.

Power of a Test

- Ideally, we should like the power of our test to be 100%
 - this is impossible because there is always a chance, albeit slim, that we could make a Type II error.
- Fortunately, however, we know which factors affect power, and thus we can control the power of a test by giving consideration to them.
- It is essential that we know the power of a proposed test at the planning stage of our investigation.
 - Clearly, we should only embark on a study if we believe that it has a 'good' chance of detecting a clinically relevant effect, if one exists (by 'good' we mean that the power should be at least 80%).
 - It is ethically irresponsible, and wasteful of time and resources, to undertake a clinical trial that has, say, only a 40% chance of detecting a real treatment effect.

Factor affecting the Power of a Test

- The sample size:
- Power increases with increasing sample size.
 - A large sample has a greater ability than a small sample to detect a clinically important effect if it exists.
 - When the sample size is very small, the test may have an inadequate power to detect a particular effect.

Factor affecting the Power of a Test

- The variability of the observations: power increases as the variability of the observations decreases
- The effect of interest: the power of the test is greater for larger effects.
 - A hypothesis test thus has a greater chance of detecting a large real effect than a small one.

Factor affecting the Power of a Test

- **The significance level:** the power is greater if the significance level is larger (this is equivalent to the probability of the Type I error (α) increasing as the probability of the Type II error (β) decreases).
 - We are more likely to detect a real effect if we decide at the planning stage that we will regard our P-value as significant if it is less than 0.05 rather than less than 0.01.
- Note that an inspection of the confidence interval for the effect of interest gives an indication of whether the power of the test was adequate.
 - A wide confidence interval results from a small sample and/or data with substantial variability, and is a suggestion of low power.

Title: Sampling Distributions of Statistics and the Central Limit Theorem

Introduction to Sampling Distributions:

- In statistics, a sampling distribution refers to the distribution of a statistic (e.g., mean, proportion) calculated from multiple samples of the same size drawn from a population.
- It provides insights into the behavior of the statistic and allows us to make inferences about the population based on sample statistics.

Key Concepts:

- **Sample Statistic:** A numerical measure calculated from a sample, such as the sample mean or sample proportion.
- **Population Parameter:** A numerical measure that describes a characteristic of the entire population, such as the population mean or population proportion.
- **Sampling Distribution:** The distribution of sample statistics obtained from multiple random samples drawn from the same population.

Characteristics of Sampling Distributions:

- Sampling distributions tend to have different properties than the population distribution.
- As the sample size increases, the sampling distribution becomes more concentrated around the population parameter.
- Sampling distributions can be used to estimate population parameters and calculate measures of uncertainty, such as standard errors.

Central Limit Theorem (CLT):

- The Central Limit Theorem is a fundamental concept in statistics.
- It states that, under certain conditions, the sampling distribution of the sample mean approximates a normal distribution regardless of the shape of the population distribution.
- The CLT applies to sample means calculated from sufficiently large, independent, and identically distributed (IID) samples.

Key Aspects of the Central Limit Theorem:

- **Sample Size:** As the sample size increases, the sampling distribution of the sample mean becomes increasingly closer to a normal distribution.
- **Independence:** The observations within the sample should be independent of each other.
- **Identically Distributed:** The data should be drawn from the same population with the same distributional characteristics.

Implications of the Central Limit Theorem:

- The CLT allows us to make important inferences about the population mean using sample means and approximate their distribution.
- It provides the basis for constructing confidence intervals and conducting hypothesis tests for population means.

- The CLT is widely applicable, making it valuable in various fields of study.

Practical Applications:

- The Central Limit Theorem is frequently used in hypothesis testing and confidence interval estimation for population means.
- It is particularly useful when the population distribution is unknown or not normally distributed.
- It enables the use of parametric statistical tests and facilitates generalizations from sample results to the population.

Understanding sampling distributions and the Central Limit Theorem is crucial for making inferential conclusions based on sample statistics. These concepts provide a foundation for statistical inference, hypothesis testing, and constructing confidence intervals. The Central Limit Theorem allows for robust and powerful statistical analysis, even when population distribution information is limited or unknown.

Title: Standard Error and Estimation

Introduction to Standard Error:

- Standard error is a measure of the precision or variability of an estimator (e.g., sample mean, sample proportion) in estimating a population parameter.
- It quantifies the average amount of uncertainty or sampling variation associated with the estimator – it is the standard deviation of a statistic.

Calculation of Standard Error:

The standard error is typically calculated using the standard deviation of the sample statistic and the sample size.

- For the sample mean, the standard error (SE) is given by $SE = (\text{standard deviation} / \sqrt{n})$, where n is the sample size.
- For other estimators, such as proportions or regression coefficients, the formula for standard error may differ but follows a similar concept.

Interpretation of Standard Error:

- A smaller standard error indicates less variability or more precision in the estimator, suggesting a more reliable estimate of the population parameter.
- A larger standard error indicates more variability or less precision in the estimator, implying increased uncertainty in estimating the population parameter.

Estimation using Standard Error:

Standard error is crucial for constructing confidence intervals and conducting hypothesis tests.

- **Confidence Intervals:** Standard error is used to calculate the margin of error in constructing confidence intervals. For example, a 95% confidence interval for the population mean is typically estimated as $(\text{sample mean} \pm 1.96 * SE)$, where 1.96 represents the critical value from the standard normal distribution for a 95% confidence level.
- **Hypothesis Testing:** Standard error helps in determining the test statistic and calculating the p-value. It is utilized to assess whether the observed difference between sample statistics and population parameters is statistically significant.

Examples from Biological Scenarios:

Example 1: Estimating Mean Body Weight:

Researchers collect a random sample of body weights (in kilograms) from a population of animals. The sample mean is 15 kg, and the sample standard deviation is 2 kg. The standard error of the sample mean can be calculated as $SE = (2 / \sqrt{n})$, where n is the sample size.

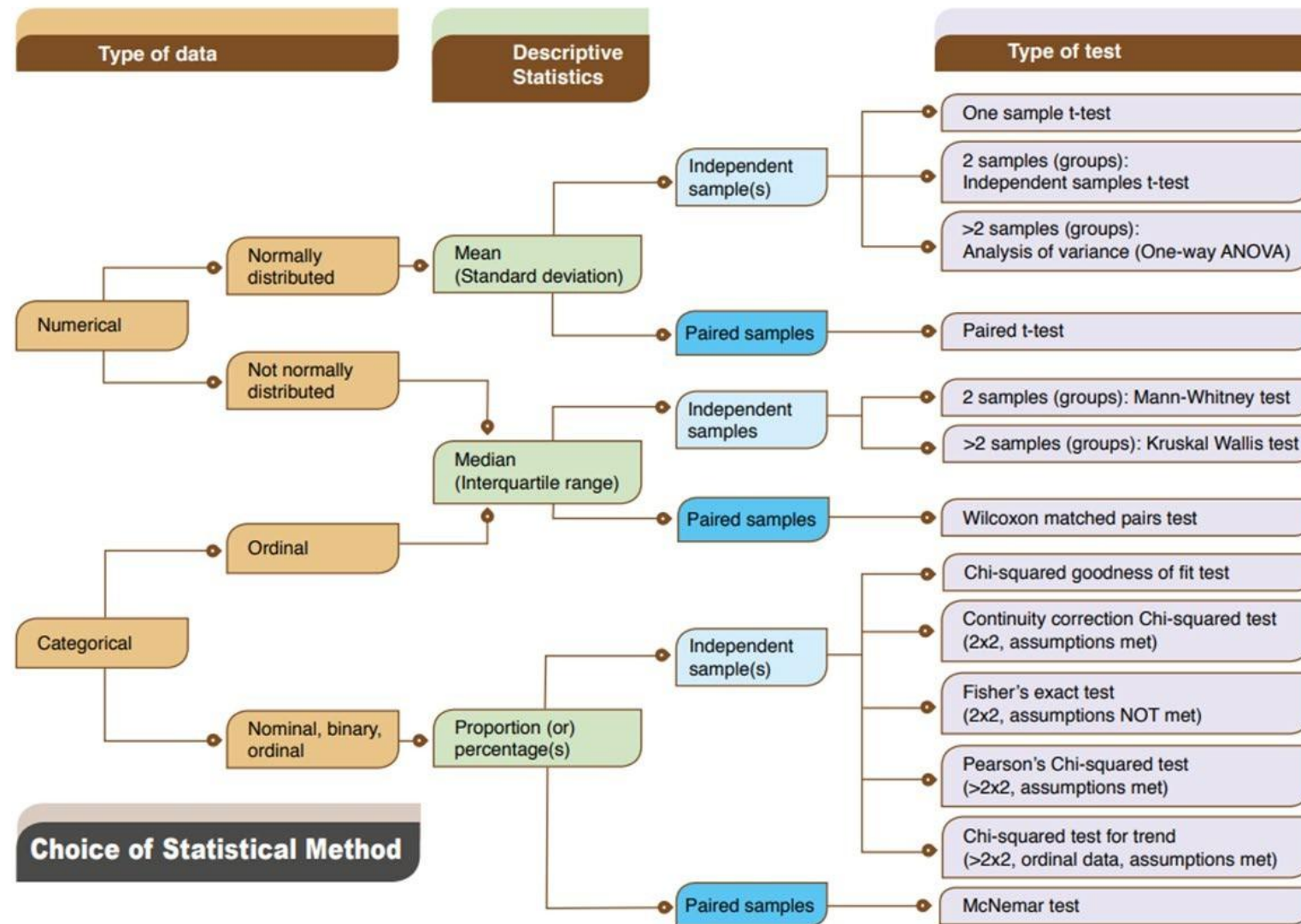
With a larger sample size, the standard error decreases, indicating a more precise estimate of the population mean. A smaller standard error suggests that the sample mean is likely to be close to the population mean.

Example 2: Estimating Proportion of Disease:

In a study investigating disease prevalence in a population, a random sample of individuals is selected, and the proportion of individuals with the disease is determined. The sample proportion is 0.25, and the sample size is 200. The standard error of the sample proportion can be calculated using the formula $SE = \sqrt{[(p(1-p)) / n]}$, where p is the sample proportion and n is the sample size. The standard error reflects the precision of the estimated disease prevalence. A smaller standard error implies a more precise estimate of the population proportion, indicating a narrower range of possible values.

Understanding standard error is essential in assessing the reliability and precision of estimators in biological scenarios. It enables researchers to construct meaningful confidence intervals, make accurate inferences, and determine the statistical significance of findings. By accounting for sampling variation, standard error provides valuable insights into the uncertainty associated with estimators and enhances the validity of statistical analyses in biological research.

Choice of Statistical Test



Module 12: Clinical Research, Statistical Analysis & Interpretation

Module Assignment

Source notes cover statistical inference, hypothesis testing, power and sample-size determination. The course outline also identifies case-control, cross-sectional, prospective and clinical-trial studies. Dedicated full notes on all study designs and complete multivariable analysis are not present.

1. Choose a clinical research question and identify the most appropriate study design: cross-sectional, case-control, cohort/prospective or clinical trial.
2. Define the exposure/intervention, outcome and key confounders.
3. State the primary statistical analysis and justify it from the variable types and study design.
4. Specify the main estimate to report and its 95% confidence interval.
5. Using a statistical output table, write a 200–300 word Results section without overstating causality.
6. List three common statistical mistakes that could lead to a misleading clinical conclusion.

Submission guidance: Show working where calculations are required. Interpret statistical results in clinical language. Use Excel, Jamovi or SPSS where appropriate.